Supplementary Appendix: Weighted Synthetic Difference-in-Differences

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This supplement accompanies "Weighted Synthetic Difference-in-Differences" (henceforth, the *main paper*) and contains technical material that supports but is not central to the headline contribution. Section A recaps notation and the standard SDID estimator. Section B contains formal statements and proofs of Propositions 1–3 from the main paper, together with discussion of when the underlying regularity conditions fail. Section C documents the bootstrap, jackknife, and placebo variance estimators with weight renormalization. Section D documents the implementation of the Monte Carlo study whose results appear in Section 4 of the main paper. Section E presents additional empirical results for the ACA application: synthetic-control results, diagnostics, and event studies, control-unit weight diagnostics, leave-one-out sensitivity, and a public-data reproduction of the double-lasso propensity score (DLPS) benchmark. Section F compares alternative strategies for incorporating treated-unit weights, explains why we adopt the modified-collapsed-form approach, and reports a Monte Carlo comparison of the three. Section G outlines extensions, including staggered adoption, time-varying period weights, and clustered standard errors.

A Setup and Notation

We restate notation from the main paper so that this supplement is self-contained. Consider a balanced panel with N units observed over T periods. The first N_0 units are untreated controls; the remaining $N_1 = N - N_0$ are treated beginning at period $T_0 + 1$, with $T_1 = T - T_0$ post-treatment periods. Let $W_{it} = \mathbf{1}\{i > N_0\} \cdot \mathbf{1}\{t > T_0\}$ denote the treatment indicator and

$$Y_{it} = L_{it} + \tau_{it} W_{it} + \epsilon_{it}$$

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the observed outcome, where L_{it} is the (untreated) potential outcome under the factor model

$$L_{it} = U_i^\top V_t + \alpha_i + \beta_t,$$

with $U_i, V_t \in \mathbb{R}^r$ low-dimensional factor loadings and α_i, β_t scalar fixed effects. Errors ϵ_{it} are assumed sub-Gaussian with bounded variance, conditional on the factor structure.

The standard SDID estimator (Arkhangelsky et al., 2021) is the bilinear form

$$\hat{\tau}^{\text{sdid}} = \begin{pmatrix} -\hat{\omega} \\ \frac{1}{N_1}\mathbf{1}_{N_1} \end{pmatrix}^\top Y \begin{pmatrix} -\hat{\lambda} \\ \frac{1}{T_1}\mathbf{1}_{T_1} \end{pmatrix},$$

with $\hat{\omega} \in \Delta^{N_0}$ and $\hat{\lambda} \in \Delta^{T_0}$ obtained by Frank-Wolfe optimization on the unit simplex against the equally-weighted treated target. The weighted SDID estimator of the main paper replaces the uniform $\frac{1}{N_1}\mathbf{1}_{N_1}$ in both the optimization target and the bilinear form with researcher-specified weights $\tilde{\omega} \in \Delta^{N_1}$:

$$\hat{\tau}^w = \begin{pmatrix} -\hat{\omega}^w \\ \tilde{\omega} \end{pmatrix}^\top Y \begin{pmatrix} -\hat{\lambda}^w \\ \frac{1}{T_1}\mathbf{1}_{T_1} \end{pmatrix}, \tag{1}$$

targeting the weighted average treatment effect

$$\tau^{(\tilde{\omega})} = \sum_{j=1}^{N_1} \tilde{\omega}_j \, \bar{\tau}_j, \quad \bar{\tau}_j = T_1^{-1} \sum_{t=T_0+1}^{T} \tau_{N_0+j,t}.$$

We collect the following assumptions, which mirror Arkhangelsky et al. (2021) with the addition of weight regularity conditions.

Assumption A1 (Outcome model). *$Y_{it} = L_{it} + \tau_{it} W_{it} + \epsilon_{it}$ with $L_{it} = U_i^\top V_t + \alpha_i + \beta_t$, where rank$(UV^\top) = r$ is fixed.*

Assumption A2 (Errors). *The error rows $\epsilon_{i\cdot} = (\epsilon_{i1}, \ldots, \epsilon_{iT})$ are independent across units i. For each unit, $\mathbb{E}[\epsilon_{it} \mid U, V, \alpha, \beta] = 0$, the errors are sub-Gaussian with parameter σ uniformly in i, t, and they satisfy a weak-dependence condition over time (e.g., a stationary mixing process across t for each i, with mixing coefficients summable in T).*

Cross-unit independence is used in the central limit step of Proposition 2; it is relaxed to independence across treatment-assignment clusters (with arbitrary within-cluster dependence) in Proposition 3.

Assumption A3 (Factor structure). *Singular values of $UV^\top$ satisfy decay conditions sufficient* *for control-trajectory matching:* $\|U^\top U/N_0\|, \|V^\top V/T_0\|$ *are bounded above and below, and the* r*-th singular value is bounded away from zero.*

Assumption A4 (Sample-size scaling). $N_0, T_0 \to \infty$ *with* $\log N_0 = o(T_0)$ *and* $\log T_0 = o(N_0)$.

These are the minimal growth conditions under which the weight-estimation error bounds invoked below are meaningful; the primitive rate conditions of Arkhangelsky et al. (2021) (those of their Theorem 1) are sufficient for them and for the inherited lemmas we cite.

Assumption A5 (Oracle weights). *There exist an intercept* $\omega_0^* \in \mathbb{R}$ *and weights* $\omega^* \in \Delta^{N_0}$ *matching the weighted treated factor structure,*

$$\sum_{j=1}^{N_1} \tilde{\omega}_j U_{N_0+j} = \sum_{i=1}^{N_0} \omega_i^* U_i, \qquad \sum_{j=1}^{N_1} \tilde{\omega}_j \alpha_{N_0+j} = \omega_0^* + \sum_{i=1}^{N_0} \omega_i^* \alpha_i,$$

and an intercept $\lambda_0^* \in \mathbb{R}$ *and weights* $\lambda^* \in \Delta^{T_0}$ *matching the post-treatment factor average,*

$$T_1^{-1} \sum_{t>T_0} V_t = \sum_{s \le T_0} \lambda_s^* V_s, \qquad T_1^{-1} \sum_{t>T_0} \beta_t = \lambda_0^* + \sum_{s \le T_0} \lambda_s^* \beta_s,$$

with $\|\omega^*\|_2 \to 0$ *as* $N_0 \to \infty$.

Because A5 imposes the matching on the factor loadings rather than period by period on L, it implies $\sum_j \tilde{\omega}_j L_{N_0+j,t} = \omega_0^* + \sum_i \omega_i^* L_{it}$ for *all* t (pre- and post-treatment) and $T_1^{-1} \sum_{t>T_0} L_{it} = \lambda_0^* + \sum_{s \le T_0} \lambda_s^* L_{is}$ for all $i \le N_0$. Three remarks. First, A5 assumes a convex (simplex) representation directly: membership of the weighted treated loadings in the linear *span* of the control loadings would not by itself deliver weights in Δ^{N_0}. Second, exact matching can be relaxed to approximate matching with vanishing error at the cost of carrying an additional bias term through the proofs, exactly as in Arkhangelsky et al. (2021). Third, the diffuseness requirement $\|\omega^*\|_2 \to 0$ rules out oracle weights concentrated on finitely many control units; it is this property that the ridge regularization in the unit-weight program transfers to $\hat{\omega}^w$.

Assumption W1 (Probability weights). $\tilde{\omega}_j \ge 0$ *and* $\sum_{j=1}^{N_1} \tilde{\omega}_j = 1$.

Assumption W2 (Effective sample size). $\max_j \tilde{\omega}_j \to 0$ *as* $N_1 \to \infty$, *equivalently* $N_1^{\mathit{eff}} := \left(\sum_j \tilde{\omega}_j^2\right)^{-1} \to \infty$.

Assumption W3 (Lindeberg condition). $\dfrac{\max_j \tilde{\omega}_j^2}{\sum_k \tilde{\omega}_k^2} \to 0$ *as* $N_1 \to \infty$.

W2 implies that no single treated unit dominates the weighted average; W3 strengthens this to a Lindeberg-type condition required for the central limit theorem applied to the weighted treated noise. W2 does not imply W3 in general (e.g., Zipf-distributed weights satisfy W2 but can violate W3); we therefore impose them separately and require W3 only for asymptotic normality.

B Proofs of Propositions 1–3

This appendix contains proofs of the three asymptotic results stated in Section 2.3 of the main paper, plus a preliminary reduction result (Proposition 0) showing that the weighted estimator nests standard SDID. Throughout, we use $C, C_1, C_2, \ldots$ to denote universal constants whose value may change line to line.

B.1 Reduction to Standard SDID

Proposition 0 (Reduction). *When $\tilde{\omega}_j = 1/N_1$ for all j, the weighted SDID estimator $\hat{\tau}^w$ equals the standard SDID estimator $\hat{\tau}^{sdid}$.*

Proof. Under uniform weights, the weighted collapsed form

$$\tilde{Y}^w_{N_0+1,t} = \sum_{j=1}^{N_1} \tilde{\omega}_j Y_{N_0+j,t} = \frac{1}{N_1} \sum_{j=1}^{N_1} Y_{N_0+j,t}$$

equals the standard collapsed form $\tilde{Y}_{N_0+1,t}$ for all $t \le T_0$, and the last column is unchanged by construction. Since the Frank-Wolfe optimization for $(\hat{\omega}^w, \hat{\lambda}^w)$ depends on the panel only through the collapsed form, $(\hat{\omega}^w, \hat{\lambda}^w) = (\hat{\omega}, \hat{\lambda})$. Substituting into the weighted estimator,

$$\hat{\tau}^w = \begin{pmatrix} -\hat{\omega} \\ \frac{1}{N_1}\mathbf{1}_{N_1} \end{pmatrix}^\top Y \begin{pmatrix} -\hat{\lambda} \\ \frac{1}{T_1}\mathbf{1}_{T_1} \end{pmatrix} = \hat{\tau}^{\mathrm{sdid}}. \quad \square$$

B.2 Consistency

Proposition 1 (Consistency). *Under Assumptions A1–A5 and W1–W2, $\hat{\tau}^w \xrightarrow{p} \tau^{(\tilde{\omega})}$ as $N_0, N_1, T_0 \to \infty$.*

Proof. We follow the decomposition strategy of Arkhangelsky et al. (2021), Theorem 1, with the substitution $\frac{1}{N_1}\mathbf{1}_{N_1} \mapsto \tilde{\omega}$ throughout. For any time weights $\lambda \in \Delta^{T_0}$ define the unit-level noise *contrast* $\xi_i(\lambda) := T_1^{-1} \sum_{t>T_0} \epsilon_{it} - \sum_{s \le T_0} \lambda_s \epsilon_{is}$, and write $\xi_i := \xi_i(\lambda^*)$.

Step 1 (Exact decomposition). Substitute $Y = L + \tau W + \epsilon$ into the bilinear form (1). The τW part returns $\tau^{(\tilde{\omega})}$ exactly. The systematic part is $f(\hat{\omega}^w, \hat{\lambda}^w)$, where

$$f(\omega,\lambda) := \sum_j \tilde{\omega}_j \Big(T_1^{-1} \textstyle\sum_{t>T_0} L_{N_0+j,t} - \sum_{s\le T_0} \lambda_s L_{N_0+j,s}\Big) - \sum_i \omega_i \Big(T_1^{-1} \sum_{t>T_0} L_{it} - \sum_{s\le T_0} \lambda_s L_{is}\Big)$$

is bilinear in (ω, λ). By the all-t matching implied by A5—the intercepts ω_0^*, λ_0^* cancel because every weight vector involved sums to one—we have $f(\omega^*, \lambda^*) = 0$, and both first-order deviation terms vanish as well: $f(\hat{\omega}^w, \lambda^*) - f(\omega^*, \lambda^*) = 0$ because the control-side contrast at λ^* is the constant λ_0^*, which sums to zero against $\hat{\omega}^w - \omega^*$; symmetrically for $f(\omega^*, \hat{\lambda}^w) - f(\omega^*, \lambda^*)$. Only the bilinear cross term survives. Collecting the noise terms,

$$\hat{\tau}^w - \tau^{(\tilde{\omega})} = \underbrace{\sum_{j=1}^{N_1} \tilde{\omega}_j \xi_{N_0+j} - \sum_{i=1}^{N_0} \omega_i^* \xi_i}_{E_1 \text{ (noise at oracle weights)}} + \underbrace{R_\omega + R_\lambda + R_{\omega\lambda}}_{E_2 \text{ (estimation error)}},$$

with

$$\begin{aligned} R_\omega &= -\sum_{i=1}^{N_0} \left(\hat{\omega}_i^w - \omega_i^*\right) \xi_i, \\ R_\lambda &= -\sum_{s\le T_0} \left(\hat{\lambda}_s^w - \lambda_s^*\right)\Big[\sum_j \tilde{\omega}_j \epsilon_{N_0+j,s} - \sum_i \hat{\omega}_i^w \epsilon_{is}\Big], \\ R_{\omega\lambda} &= \left(\hat{\omega}^w - \omega^*\right)^\top L_{\text{pre}} \left(\hat{\lambda}^w - \lambda^*\right), \end{aligned}$$

where L_{pre} is the $N_0 \times T_0$ control block of L.

Step 2 (Noise at oracle weights). Under A2, $\text{Var}(\xi_i) \le C\sigma^2\big(T_1^{-1} + \|\lambda^*\|_2^2 + 2T_1^{-1/2}\|\lambda^*\|_2\big) \le C'$ uniformly in i: the within-unit serial covariance between the post-period average and the λ^*-weighted pre-period sum is absorbed into $\text{Var}(\xi_i)$, which is why the contrast—rather than its post and pre pieces separately—is the right object under serially dependent errors. By cross-unit independence (A2),

$$\text{Var}\left(\sum_j \tilde{\omega}_j \xi_{N_0+j}\right) \le C' \sum_j \tilde{\omega}_j^2 = \frac{C'}{N_1^{\text{eff}}} \to 0$$

by W2, and $\text{Var}\big(\sum_i \omega_i^* \xi_i\big) \le C'\|\omega^*\|_2^2 \to 0$ by A5. Both pieces of E_1 are $o_p(1)$ by Markov's inequality.

Step 3 (Estimation error). Under A2–A4 the weight-estimation errors vanish, $\|\hat{\omega}^w - \omega^*\|_2 = o_p(1)$ and $\|\hat{\lambda}^w - \lambda^*\|_2 = o_p(1)$, with the ridge regularization in the unit-weight program transferring the diffuseness of ω^* (A5) to $\hat{\omega}^w$. These are the estimation-error arguments behind Arkhangelsky et al.

(2021)'s Theorem 1, and they apply with $\tilde{\omega}$ in place of $N_1^{-1}\mathbf{1}$: they depend on the treated block only through the target column of the collapsed form, which under W1 is a fixed convex combination of treated trajectories and, by A5, admits an oracle representation in the control block. Consequently R_ω and R_λ are $o_p(1)$, each pairing an $o_p(1)$ weight-estimation error with noise whose relevant projections have bounded variance. For the cross term, expand $L_{\text{pre}} = U_c V_{\text{pre}}^\top + \alpha_c \mathbf{1}^\top + \mathbf{1}\beta_{\text{pre}}^\top$: the two fixed-effect components vanish *exactly*, since $(\hat{\omega}^w - \omega^*)^\top \alpha_c \cdot \mathbf{1}^\top(\hat{\lambda}^w - \lambda^*) = 0$ and $\mathbf{1}^\top(\hat{\omega}^w - \omega^*) \cdot \beta_{\text{pre}}^\top(\hat{\lambda}^w - \lambda^*) = 0$ (each difference of simplex vectors sums to zero), leaving

$$|R_{\omega\lambda}| = \left|(\hat{\omega}^w - \omega^*)^\top U_c \cdot V_{\text{pre}}^\top(\hat{\lambda}^w - \lambda^*)\right| \leq \|U_c^\top(\hat{\omega}^w - \omega^*)\| \, \|V_{\text{pre}}^\top(\hat{\lambda}^w - \lambda^*)\| = o_p(1),$$

a product of loading-space estimation errors, each controlled by the same inherited arguments. (Bounding $R_{\omega\lambda}$ by $\|\hat{\omega}^w - \omega^*\|_2 \, \|L\|_{\text{op}} \, \|\hat{\lambda}^w - \lambda^*\|_2$ would *not* suffice, since $\|L\|_{\text{op}} \asymp \sqrt{N_0 T_0}$ for a low-rank matrix with bounded entries; the rank-r expansion is what makes the cross term manageable.)

Combining Steps 2–3, each component is $o_p(1)$, so $\hat{\tau}^w - \tau^{(\tilde{\omega})} \xrightarrow{p} 0$. □

B.3 Asymptotic Normality

Proposition 2 (Asymptotic normality). *Under Assumptions A1–A5 and W1–W3, and the negligibility condition*

(R) $\quad \sqrt{N_1^{\text{eff}}}\left(\hat{\tau}^w - \tau^{(\tilde{\omega})} - \sum_j \tilde{\omega}_j \xi_{N_0+j}\right) = o_p(1),$

i.e., the control-noise and weight-estimation terms of the Proposition 1 decomposition are negligible at the $\sqrt{N_1^{\text{eff}}}$ scale, we have

$$\sqrt{N_1^{\mathit{eff}}}\left(\hat{\tau}^w - \tau^{(\tilde{\omega})}\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma_\xi^2\right), \qquad \sigma_\xi^2 = \lim_{N_1 \to \infty} \frac{\sum_j \tilde{\omega}_j^2 \, \mathit{Var}(\xi_{N_0+j})}{\sum_k \tilde{\omega}_k^2},$$

where $\xi_{N_0+j} = T_1^{-1}\sum_{t>T_0} \epsilon_{N_0+j,t} - \sum_{s \leq T_0} \lambda_s^* \epsilon_{N_0+j,s}$ is the post-minus-pre noise contrast of treated unit j.

Proof.

Step 1 (CLT for the treated-noise contrast). Let $Z_n := \sqrt{N_1^{\text{eff}}}\sum_j \tilde{\omega}_j \xi_{N_0+j}$. The summands are independent across j (A2), mean zero, with variances $\sigma_j^2 := \text{Var}(\xi_{N_0+j})$ bounded above uniformly (Step 2 of the Proposition 1 proof) and assumed bounded away from zero. The Lindeberg–Feller condition for the triangular array $\{\tilde{\omega}_j \xi_{N_0+j}\}$,

$$\frac{\max_j \tilde{\omega}_j^2 \, \sigma_j^2}{\sum_k \tilde{\omega}_k^2 \, \sigma_k^2} \to 0,$$

reduces under uniformly bounded σ_j^2 to W3. Hence $Z_n \xrightarrow{d} \mathcal{N}(0, \sigma_\xi^2)$ with σ_ξ^2 as displayed. Because ξ is the post-minus-pre *contrast*, σ_ξ^2 already incorporates the within-unit serial covariance between post-period and λ^*-weighted pre-period noise; no independence between those two pieces is required (nor would it hold under serially dependent errors).

Step 2 (Negligible remainder). By (R), $\sqrt{N_1^{\text{eff}}}(\hat{\tau}^w - \tau^{(\tilde{\omega})}) = Z_n + o_p(1)$, and the conclusion follows by Slutsky's theorem. $\square$

Remark (Primitive conditions for (R)). Condition (R) has two parts. The control-noise term $\sqrt{N_1^{\text{eff}}} \sum_i \omega_i^* \xi_i$ has variance of order $N_1^{\text{eff}} \|\omega^*\|_2^2$, so it requires the oracle control weights to be more diffuse than the treated weights are concentrated ($N_1^{\text{eff}} \|\omega^*\|_2^2 \to 0$)—plausible when N_1^{eff} grows slowly relative to the effective number of oracle control units. The weight-estimation terms $R_\omega, R_\lambda, R_{\omega\lambda}$ require the estimation-error rates of Arkhangelsky et al. (2021) to dominate $\sqrt{N_1^{\text{eff}}}$; the rate conditions of their Theorem 1 play exactly this role in the unweighted case (where $N_1^{\text{eff}} = N_1$) and carry over under the same substitution as in Proposition 1. We do not restate primitive rates here: in the applications we target, the binding constraints are W2–W3 themselves, and inference proceeds by bootstrap rather than by plug-in use of σ_ξ^2.

Remark (When is (R) plausible?). Condition (R) formalizes the requirement that uncertainty from the estimated weights $(\hat{\omega}^w, \hat{\lambda}^w)$ and from the control-noise average be negligible relative to the treated-noise contrast, whose variance is of order $1/N_1^{\text{eff}}$. It is most plausible when N_1^{eff} is small relative to the control block and the pre-period is long. In our ACA application, $N_1^{\text{eff}} \approx 102$ against $N_0 = 1,643$ control counties—favorable on the cross-sectional side—but the pre-period is short ($T_0 = 5$), so the time-weight estimation error is unlikely to be negligible in the sense of (R). This is one more reason our inference relies on the cluster bootstrap—which resamples the entire estimation pipeline and therefore captures, in finite samples, the contributions that (R) treats as asymptotically negligible—together with randomization inference, rather than on plug-in asymptotic intervals.

B.4 Cluster-Robust Variance

Proposition 3 (Cluster-robust variance). *Suppose treatment is assigned at the cluster level, with K clusters indexed $k = 1, \ldots, K$ and cluster membership $g(i) \in \{1, \ldots, K\}$. Let $\tilde{\omega}^{(k)} =$*

$\sum_{j:\, g(N_0+j)=k} \tilde\omega_j$ *be the cluster-aggregated treated weight,* $\xi^{(k)} = \sum_{j:\, g(N_0+j)=k} \tilde\omega_j\, \xi_{N_0+j}$ *the cluster-aggregated noise contrast, and* $K^{\mathit{eff}} = \big(\sum_k (\tilde\omega^{(k)})^2\big)^{-1}$ *the effective number of treated clusters. Assume*

(CB1) $K \to \infty$ *and* $\max_k \tilde\omega^{(k)} \to 0$ *(no dominant cluster), with the cluster-level Lindeberg condition* $\max_k (\tilde\omega^{(k)})^2 / \sum_{k'} (\tilde\omega^{(k')})^2 \to 0$*;*

(CB2) *errors are mean-zero, with arbitrary dependence within clusters and independence across clusters (replacing the cross-unit independence in A2);*

(CB3) *the analogue of condition (R) at the cluster scale:* $\sqrt{K^{\mathit{eff}}}(\hat\tau^w - \tau^{(\tilde\omega)} - \sum_k \xi^{(k)}) = o_p(1)$*.*

Then

$$\sqrt{K^{\mathrm{eff}}}\left(\hat\tau^w - \tau^{(\tilde\omega)}\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma^2_{\mathrm{cl}}\right), \qquad \sigma^2_{\mathrm{cl}} = \lim_{K\to\infty} \frac{\sum_k \mathrm{Var}\big(\xi^{(k)}\big)}{\sum_{k'} \big(\tilde\omega^{(k')}\big)^2},$$

and the cluster bootstrap variance estimator $\widehat{V}^{cb}_w = (B-1)^{-1} \sum_b (\hat\tau^{w,(b)} - \bar\tau^w)^2$ *is consistent for* $\sigma^2_{cl}/K^{\mathit{eff}}$*, where* $\hat\tau^{w,(b)}$ *is computed on a cluster-resampled bootstrap draw with weight renormalization* $\tilde\omega^{(b)}_j = \tilde\omega_j n^{(b)}_j / \sum_{\ell=1}^{N_1} \tilde\omega_\ell n^{(b)}_\ell$ *and analogous renormalization of* $\hat\omega^w$*.*

Proof. The central limit step repeats Proposition 2 one level up: the $\xi^{(k)}$ are independent across clusters by (CB2)—within-cluster covariances among the ξ_{N_0+j} are *absorbed into* $\mathrm{Var}(\xi^{(k)})$ rather than restricted—mean zero, with the Lindeberg condition supplied by (CB1), and (CB3) disposes of the remainder. Note that σ^2_{cl} does not in general equal σ^2_ξ of Proposition 2: it exceeds it whenever noise is positively correlated within clusters, which is precisely the situation clustering is designed for. For the bootstrap step, the cluster-bootstrap consistency argument of Cameron et al. (2008) (see also Clarke et al. (2023)) applies to the bilinear form (1): (i) resampling whole clusters reproduces the cross-cluster independence structure under which the limit was derived; (ii) the renormalization preserves the convex-combination structure of $\tilde\omega$ within each draw, so each $\hat\tau^{w,(b)}$ is itself a valid weighted SDID estimate; (iii) under (CB1)–(CB3) the bootstrap distribution of $\sqrt{K^{\mathrm{eff}}}(\hat\tau^{w,(b)} - \hat\tau^w)$ converges to the same Gaussian limit. The cluster-jackknife variant follows by an analogous argument with cluster-level leave-one-out. We emphasize that no bound on cluster *sizes* is imposed—clusters may contain arbitrarily many units (states contain hundreds of counties in our application); what matters is that no cluster's *weight share* dominates. □

B.5 When the Asymptotics Fail

The conditions W2, W3, and (CB1) all impose a "no-dominant-weight" requirement at the appropriate level. They fail, or lose practical bite, when:

1. *Population weights with extreme concentration.* The Pareto tail of population sizes for

U.S. counties or states means that population-weighted estimands often have effective sample sizes far below the number of treated units. In our ACA application, $N_1^{\text{eff}} \approx 102$ relative to $N_1 = 1,181$, with a single county (Los Angeles) carrying roughly 6% of total weight. Such weights satisfy W2 in the strict mathematical sense (the largest weight does shrink as new counties are added), but the rate of convergence is too slow for finite-sample asymptotic intervals to be reliable.

2. *Single-cluster dominance.* When treatment is assigned at the state level and one or two states contain a disproportionate share of the treated population (California, Texas, New York in U.S. applications), the no-dominant-cluster condition (CB1) is strained at the cluster level. In our ACA application, California alone contains roughly 23% of treated population mass across the 26 treated jurisdictions. State-level cluster bootstrapping is nonetheless the most defensible resampling scheme available—treatment is assigned at the state level, and the remaining clusters are individually small—but a largest-cluster share of this magnitude means (CB1) is itself under pressure, which is why the application reports randomization-inference p-values alongside the cluster-bootstrap intervals rather than leaning on either alone.

3. *Adversarial weight choices.* If the researcher chooses weights that intentionally concentrate on a single unit (e.g., to estimate a unit-specific effect), the framework reduces to a single-unit synthetic control problem, for which Cattaneo et al. (2021)'s prediction-interval inference is more appropriate than our bilinear-form CLT.

In any of these settings, three remedies are available within our framework. *Cluster-robust bootstrap* at a coarser aggregation level (e.g., region rather than state) restores (CB1) at the cost of treating finer within-region heterogeneity as exchangeable. *Randomization inference* via the placebo distribution, as illustrated in the main paper's in-space placebo, provides an exact finite-sample test that does not rely on (W3) or (CB1). *Subsampling* of treated units, in the spirit of Politis et al. (1999), can give valid confidence intervals under weaker conditions but at the cost of reduced power. We recommend reporting both the bootstrap-based interval and the randomization-inference p-value when (W2)–(W3) are quantitatively suspect.

C Inference Algorithms

This appendix documents the bootstrap, jackknife, and placebo variance estimators with weight renormalization, including the cluster-robust extensions referenced in the main paper.

C.1 Bootstrap

The unit-level bootstrap of Arkhangelsky et al. (2021) resamples units with replacement. With researcher-specified treated-unit weights, in bootstrap draw b unit j appears $n_j^{(b)}$ times, and the weights are renormalized:

$$\tilde{\omega}_j^{(b)} = \frac{\tilde{\omega}_j \cdot n_j^{(b)}}{\sum_{\ell=1}^{N_1} \tilde{\omega}_\ell \cdot n_\ell^{(b)}}.$$

Control weights are renormalized analogously when control units are resampled. Draws in which no treated units appear are discarded. Because the renormalized weights vary across draws, each replication estimates a slightly perturbed weighted estimand; this is standard for weighted estimators and is part of the sampling variability the bootstrap is designed to capture. The bootstrap variance is

$$\widehat{V}_w^{\text{boot}} = \frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\tau}^{w,(b)} - \bar{\tau}^{w} \right)^2.$$

The cluster bootstrap of Clarke et al. (2023) generalizes by resampling clusters (rather than individual units) with replacement. The renormalization step then operates at both the unit and cluster levels: when cluster k appears $m_k^{(b)}$ times, all units within k inherit the multiplier $m_k^{(b)}$ before the unit-level renormalization is applied.

C.2 Jackknife

The leave-one-out jackknife drops unit i in turn. When the dropped unit is a treated unit j, the remaining treated weights are renormalized:

$$\tilde{\omega}_k^{(-j)} = \frac{\tilde{\omega}_k}{\sum_{\ell \neq j} \tilde{\omega}_\ell}, \quad k \neq j.$$

We use the *fixed-weight* jackknife, holding $\hat{\omega}^w$ and $\hat{\lambda}^w$ at their full-sample values while renormalizing only the treated and control weight vectors. The jackknife variance is

$$\widehat{V}_w^{\text{jk}} = \frac{N-1}{N} \sum_{i=1}^{N} \left(\hat{\tau}^{w,(-i)} - \bar{\tau}^{w,(\cdot)} \right)^2.$$

Following Arkhangelsky et al. (2021), the jackknife is not recommended for the SC variant due to the non-differentiability of the SC weights at the boundary of the simplex.

C.3 Placebo

The placebo method assigns N_1 control units as pseudo-treated and estimates the treatment effect on them, repeating B times. Since the original $\tilde{\omega}$ does not naturally map to control units, the placebo variance estimator must choose a pseudo-treated weight scheme. We provide three options in the software:

- `"uniform"` (default): pseudo-treated units receive equal weights $1/N_1$. This is the cleanest analog of the standard placebo and tests the sharp null of no treatment effect under symmetric weighting.
- `"size_match"`: pseudo-treated units are weighted by their pre-treatment outcome levels (as a proxy for unit size when explicit size variables are unavailable). This preserves the *concentration* of the original $\tilde{\omega}$ but not its identity.
- `"permute"`: the original $\tilde{\omega}$ vector is randomly permuted across the N_1 pseudo-treated units. This preserves both the marginal distribution of weights and (in expectation) their relationship to control-unit characteristics.

The cluster-robust placebo extension is left to future work; in the meantime, randomization inference via the in-space placebo distribution (as illustrated in the main paper) provides an alternative test that conditions on the realized weight structure.

D Monte Carlo Implementation Details

The Monte Carlo design and headline results—bias, RMSE, and 95% confidence-interval coverage of the equally-weighted and population-weighted SDID estimators across 18 grid configurations plus an ACA-calibrated cell matching the application's panel dimensions and weight concentration—appear in Section 4 of the main paper. This appendix documents the implementation. The simulation sweep is run outside the manuscript build by `scripts/run_mc_simulations.R`, which parallelizes the (configuration, simulation) grid across cores with batched checkpointing and resume support; each task is seeded as `set.seed(20240101 + sim)`, so results are identical to the equivalent sequential loop. The script writes `paper/data/mc_results.csv`, which the main paper's tables read. (The simulation count of 100 per configuration and the 50 bootstrap/placebo replications are smaller than would be ideal for tightly estimated coverage; we view the results as illustrative of the qualitative pattern rather than as precise coverage estimates.) The simulation functions are reproduced below for reference.

E Additional Empirical Results

This appendix contains additional empirical results for the ACA application.

E.1 Synthetic Control Results and Diagnostics

Earlier drafts of the main paper reported synthetic control (SC) estimates alongside DID and SDID in the main results table (main-paper Table 2). The SC estimates are an extreme outlier relative to the DID–SDID family, and this appendix reports them together with the standard battery of post-estimation diagnostics—pre-treatment fit, leave-one-out donor sensitivity, and in-time placebos—to document why their magnitude differs.

As implemented here (following Arkhangelsky et al. (2021)), SC sets the time weights to zero and omits the unit intercept: the estimator compares post-treatment *levels* of the (weighted) treated average against a convex combination of control-county levels, with no allowance for a permanent level difference between the treated average and its synthetic control. Any level gap that the simplex-constrained match cannot close therefore loads directly into the post-treatment difference and is read as a treatment effect. DID and SDID, by contrast, difference out unit-specific levels.

Table 1: Synthetic control estimates of the 2014 Medicaid expansion effect.

Weighting	Estimate	SE (boot)	SE (cluster)
Equally weighted	-92.08	39.95	49.59
Population weighted	-97.43	35.82	43.09

Notes:
Deaths per 100,000, ages 20–64. SC estimated as in Arkhangelsky et al. (2021): zero time weights, no unit intercept in the control match. SE (boot) is the county-level bootstrap and SE (cluster) the state-clustered bootstrap, both with 500 replications, matching the standard-error conventions of main-paper Table 2.

Table 2: Pre-treatment fit and donor-concentration diagnostics, SC vs. SDID.

Model	Pre-RMSPE (level)	Pre-RMSPE (intercept-adj.)	Mean pre-period gap	Effective N_0	Top-10 $\hat{\omega}$ share
SC, equally weighted	0.00	0.00	0.00	4.07	1.00
SC, population weighted	0.00	0.00	0.00	5.08	1.00
SDID, equally weighted	43.16	1.38	-43.13	1628.71	0.01
SDID, population weighted	127.69	1.57	-127.68	1622.64	0.01

Notes:
Pre-RMSPE (level) is the root mean squared gap between the weighted treated trajectory and the synthetic-control trajectory over pre-treatment years; the intercept-adjusted version nets out the mean pre-period gap and is the fit measure relevant for SDID, which permits level differences. The mean pre-period gap is the level difference between the weighted treated trajectory and the synthetic control; SDID absorbs it through the unit intercept (hence its large raw gap), whereas SC, lacking an intercept, drives it to zero through the donor weights—at the cost of a very low effective N_0. Effective N_0 is the Kish effective number of donor counties $(\sum_i \hat{\omega}_i^2)^{-1}$; the last column is the share of donor weight on the ten largest-$\hat{\omega}$ counties.

Table 1 reports the SC estimates under both weightings with the same dual standard errors as main-paper Table 2: the equally-weighted SC estimate is -92.08 and the population-weighted estimate -97.43 deaths per 100,000, both substantially larger in magnitude than the corresponding SDID estimates of -0.71 and -17.45. The fit diagnostics in the second table show why: the mechanism is donor concentration, not an unclosed level gap. Because SC uses no time weights and no unit intercept, it matches the pre-treatment treated trajectory in levels by loading almost all weight onto a handful of donors: the effective number of donor counties is just 5 (population weighted), with roughly 100% of the weight on its ten largest donors. So sparse a donor set interpolates the short pre-period almost exactly (level pre-RMSPE 0.00), but the match does not generalize: post-treatment the few donors diverge from the treated average, and with no intercept to absorb the difference the divergence is read directly as an enormous effect. SDID, by contrast, spreads weight across roughly 1,623 effective donors and permits a unit intercept: it carries a large *raw* pre-period level gap (-127.68, absorbed by the intercept) yet fits well once the intercept is netted out (intercept-adjusted pre-RMSPE 1.57). The SC outlier is therefore a symptom of sparse, unregularized level matching on a short pre-period, which is why we report it only here and exclude it from the main-paper comparison.

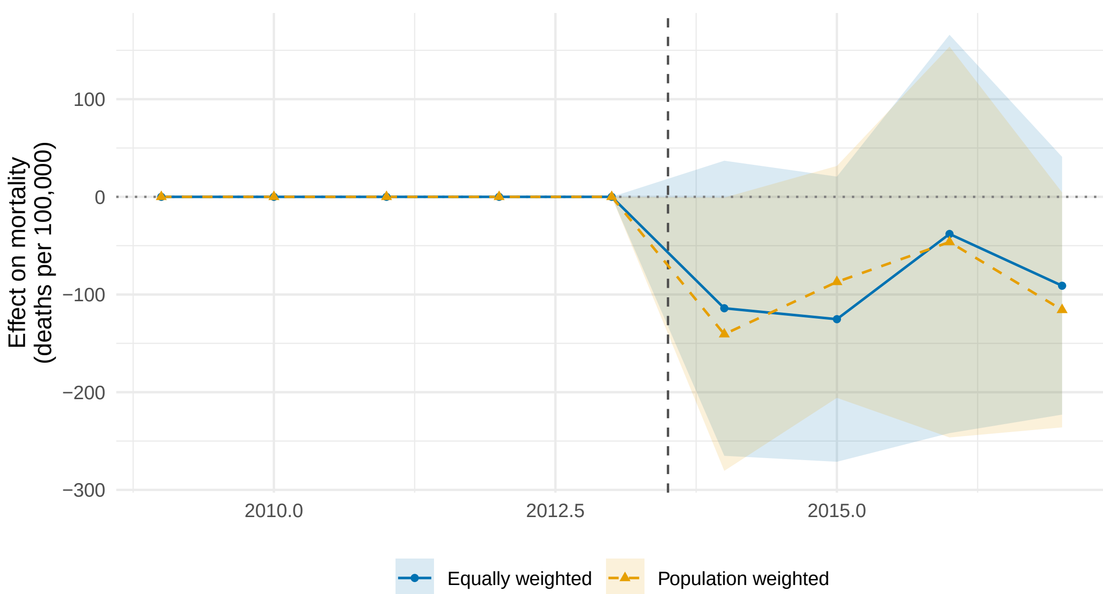

Figure 1: Synthetic-control event study.
Notes: Year-specific SC effects under equal (blue circles) and population (orange triangles) weighting of treated counties; shaded bands are 95% state-clustered bootstrap confidence intervals (500 replications). The vertical dashed line marks treatment onset (2014). SC matches pre-treatment levels by construction, so pre-period coefficients sit near zero; the population-weighted post-treatment path diverges sharply, reflecting the donor-concentration over-fit documented above.

Figure E.1 plots the SC event-study path under both weightings. Because SC matches pre-treatment *levels* with no time weights or intercept, the pre-treatment coefficients are pinned near zero by construction—a near-perfect in-sample fit on the handful of donors it selects. Post-treatment, the equally-weighted path stays close to zero while the population-weighted path drops steeply to the -97.43 reported in Table 1: the sparse donor set that fit the pre-period does not track the (population-weighted) treated average after 2014, and with no intercept to absorb the resulting gap the divergence is read entirely as a treatment effect. The event study thus visualizes the over-fit diagnosed above, and motivates excluding SC from the main-paper comparison.

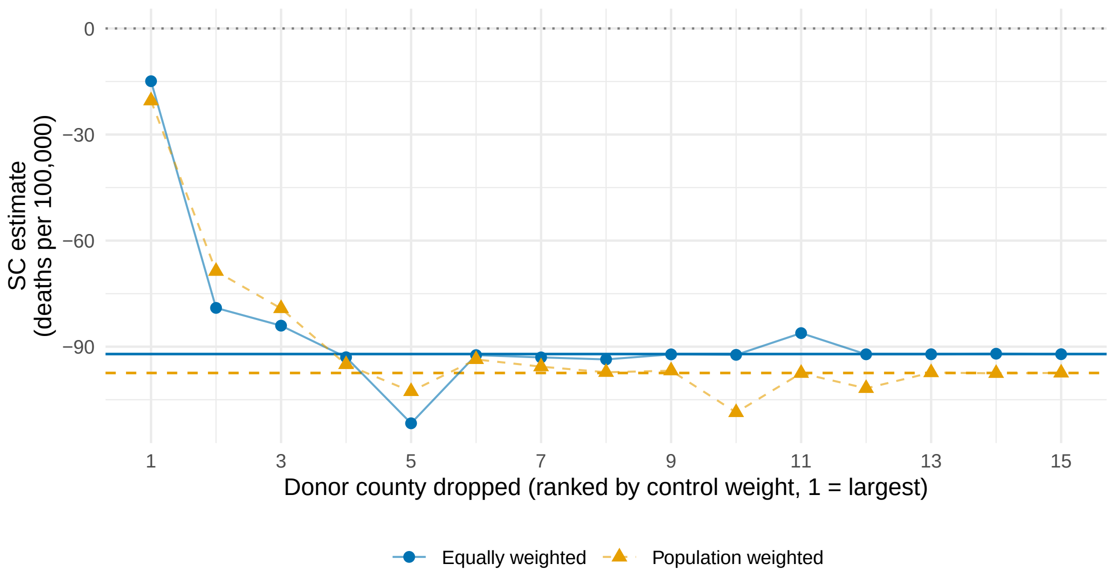

Figure 2: Leave-one-out donor sensitivity of the SC estimates.
Notes: Each point re-estimates the synthetic control after dropping one of the 15 largest-weight donor (control) counties, ranked by $\hat{\omega}$ within each weighting scheme. Horizontal lines mark the full-sample SC estimates (solid: equally weighted; dashed: population weighted); the dotted line marks zero. Circles: equally weighted; triangles: population weighted.

Table 3: In-time placebo tests for the SC estimator.

	Equally weighted		Population weighted	
Placebo year	Eq. estimate	Eq. SE	Pop. estimate	Pop. SE
2011	169.36	99.27	101.41	97.74
2012	152.52	87.82	91.27	75.14

Notes:
Sample restricted to pre-treatment years (2009–2013) with simulated treatment onset in the placebo year; no policy change occurred, so a well-behaved estimator should return zero. SEs are state-clustered bootstrap (200 replications).

The leave-one-out exercise shows that the SC estimates are sensitive to the donor pool: dropping individual large-weight donors moves the estimate materially. The in-time placebos are the more diagnostic test: where the main paper's SDID in-time placebos are at most a few deaths per 100,000 (indistinguishable from zero equally weighted, around -7.5 population weighted—itself a bias signal the main paper discusses in Section 3.6), the SC placebo estimates of 101.41 (2011) and 91.27 (2012, population weighted) are an order of magnitude larger, indicating how readily SC's sparse, over-fit donor match generates a spurious effect in years when no treatment occurred. Taken together, the diagnostics attribute the outsized SC estimates to the estimator's sparse, un-regularized level matching rather than to information about the policy: SC answers a different question than DID and SDID in this application, and we therefore exclude it from the main-paper comparison.

E.2 Control Unit Weight Diagnostics

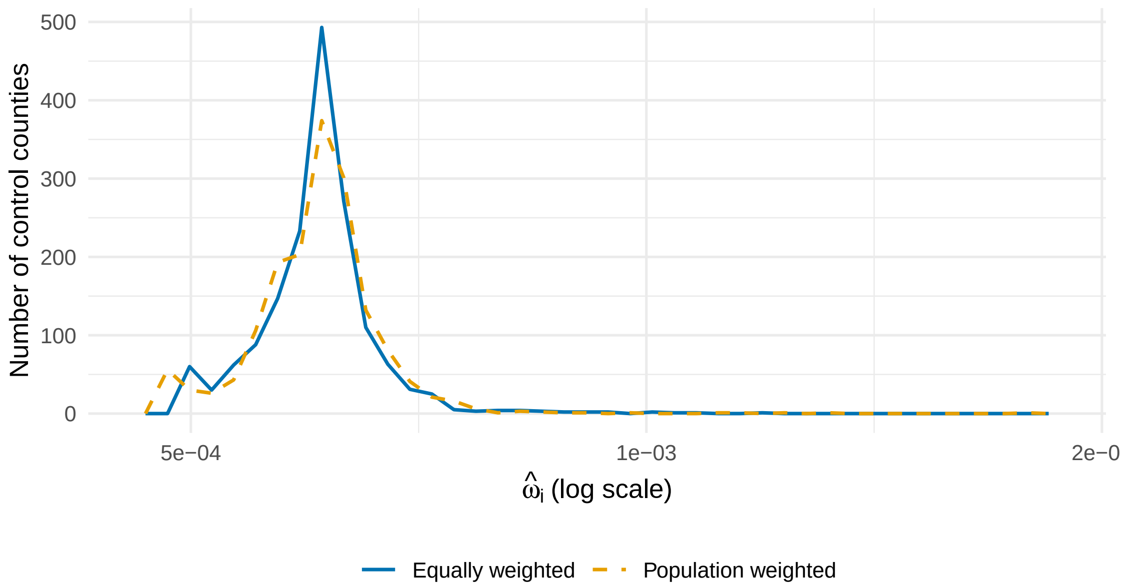

Figure 3: Distribution of SDID control-unit weights.
Notes: Frequency polygons of the estimated control weights $\hat{\omega}_i$ under equal (solid) vs. population (dashed) weighting of treated counties. Counties with zero weight excluded; the x-axis is on a log scale.

Figure E.3 shows the distribution of control unit weights under both specifications. The equally-weighted SDID places nonzero weight on 1643 control counties (effective $N_0 = 1628.7$); the population-weighted SDID uses 1643 counties (effective $N_0 = 1622.6$). Both spread weight broadly across the control pool, with similar effective sample sizes, reflecting the regularization built into the SDID algorithm. The *identity* of which controls receive the most weight shifts: under population weighting, the Frank-Wolfe algorithm matches the population-weighted treated trajectory, which is dominated by large urban counties, pulling the synthetic control toward urban-similar control counties.

E.3 Leave-One-Out Sensitivity

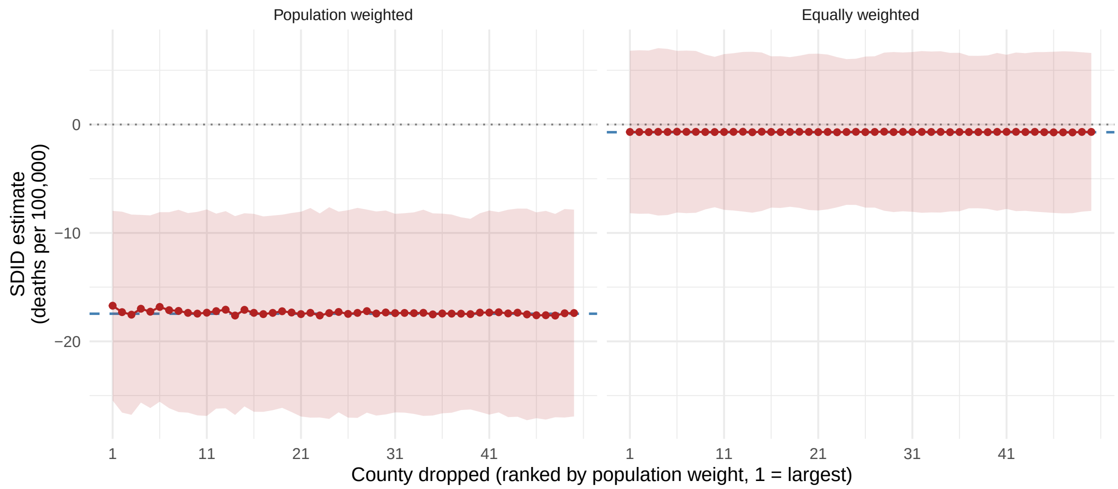

Figure 4: Leave-one-out sensitivity of the SDID estimate, by weighting.
Notes: Each point re-estimates SDID after dropping one of the 50 largest treated counties by population weight (1 = largest), under population weighting (left) and equal weighting (right). Dashed blue lines are the full-sample baselines; shaded bands are 95% state-clustered bootstrap confidence intervals (500 replications). The shared vertical scale shows both estimands are individually stable yet far apart.

Figure E.4 reports the leave-one-out sensitivity under both weightings. The population-weighted estimate clusters tightly around its baseline across all 50 counties (ranging from -17.6 to -16.7 deaths per 100,000) and remains negative and statistically distinguishable from zero throughout; even Los Angeles County (the largest weight, $\approx 6\%$ of total) does not materially shift it when removed. The equally-weighted estimate is likewise stable, hovering near zero (-0.7 to -0.7). Neither estimand is driven by any single county—each is individually robust, yet they sit far apart. This is the leave-one-out analogue of the main-paper point: the gap between the two estimates is a property of the *weighting* (a broadly negative county-size gradient that population weighting emphasizes), not of one or two influential counties.

E.4 Reproducing BV2020 with Public Data

This appendix documents our public-data reproduction of the all-cause mortality results in Borgschulte and Vogler (2020) (BV): the headline -11.36 deaths per 100,000 estimate (their Table 2, Panel A) and the heterogeneity by baseline uninsurance (their columns 10–11). BV use restricted-access NVSS mortality microdata; we use public CDC WONDER tabulations, recovering the county-year cells that WONDER suppresses (counts of 1–9) by allocating the exact state-year residual—the state total, which is never suppressed, minus the sum of unsuppressed county counts—across suppressed cells in proportion to working-age population, clamped to the feasible $[1, 9]$ range.

The resulting estimation sample is essentially BV's: within two counties of their 2,823-county pre-trim sample (exact counts below). The exercise is purely diagnostic: it is *not* the source of the comparison shown in main-paper Table 4, which uses BV's published value directly.

E.4.1 Covariate menu

We assemble BV's propensity-score candidate menu (their footnote 8 and Appendix Table A.2) from public sources, documented in `scripts/pull_bv_covariates.R` and `scripts/pull_county_votes.R`:

- Five working-age population shares (20–24, 25–34, 35–44, 45–54, 55–64; shares of the 20–64 population), percent male, percent white, percent Black, percent Hispanic, poverty rate, log median household income, and county population density from the ACS 2009–2013 5-year file and the 2010 county shapefile;
- Pre-expansion uninsured rate for ages 18–64 from Census SAHIE 2013;
- All-cause mortality rates for each year 2005–2009 from the analysis panel and its 2005–2008 extension, matching BV's design of holding out the 2010–2013 outcome years from the propensity score;
- County-level Obama vote shares in 2008 and 2012 from the MIT Election Lab county presidential returns;
- Democratic-governor 2010 indicator coded as the party of the sitting governor at ACA passage (March 2010), from National Governors Association records;
- SEER county population by age and sex, and LAUS county unemployment rates, for the time-varying difference-in-differences controls.

The one BV candidate we omit is logged state health and welfare expenditures 2005–2013: BV's own selection procedure never retains it (their Appendix Table A.2), so the omission cannot affect the estimated propensity score model meaningfully.

E.4.2 Estimation procedure

Estimation proceeds in five steps following BV: (i) collapse the panel to one row per county and average the time-varying covariates over 2009–2013; (ii) select predictors by double lasso (Belloni et al., 2014)—a square-root lasso of pre-expansion (2005–2009) mortality on the candidates and a square-root lasso of the expansion indicator on the candidates, taking the union of the selected sets, as in BV's footnote 8; (iii) fit a logistic regression of expansion on the union to construct the propensity score; (iv) trim counties with $\hat{p} < 0.038$ or $\hat{p} > 0.971$ (the BV trim, in propensity-score units, not quantile units); (v) form inverse-probability weights ipw $= D + (1 - D) \cdot \hat{p}/(1 - \hat{p})$, multiply by the county population aged 20–64, and estimate the weighted DID with two-way (county, year) fixed effects and state-clustered standard errors. The selected covariates closely mirror BV's published selection: demographics, income, unemployment, the uninsured rate, both Obama vote shares, the Democratic-governor indicator, and the five yearly pre-expansion mortality rates.

E.4.3 Reproduction result

Table 4: Public-data reproduction of Borgschulte and Vogler (2020), Table 2 Panel A.

Specification	Ours	BV (2020)
(1) Base (no controls)	−9.56 (5.05)	−14.83 (6.12)
(2) With controls	−6.62 (4.14)	−11.36 (3.59)
(10) High-uninsured counties	−13.76 (3.58)	−12.40 (4.22)
(11) Low-uninsured counties	−3.49 (4.62)	−3.96 (4.42)

Notes:
All-cause mortality, deaths per 100,000; state-clustered standard errors in parentheses. Column specifications follow BV: (1) propensity-score-weighted DID with county and year fixed effects only; (2) adds the six panel-lasso controls from BV's table notes; (10)/(11) split the trimmed sample at the median pre-expansion uninsured rate. Our estimates apply BV's procedure—square-root-lasso double selection, logit propensity score, trim at $[0.038, 0.971]$, working-age-population $\times$ IPW weights—to the public CDC WONDER panel with suppression-imputed small cells.

The double lasso selects 18 of 26 candidates, closely matching the selection BV report in their Appendix Table A.2 (both include the demographic shares, income, unemployment, the uninsured rate, both Obama vote shares, the Democratic-governor indicator, and all five yearly pre-expansion mortality rates). The complete-case sample is 2,821 counties against BV's 2,823; the propensity-score trim removes 327 counties against their 563.

Table 4 reports the reproduction. The structure of BV's results reproduces well on public data: every specification is negative; the base estimate of -9.56 (SE 5.05) is within one standard error of BV's -14.83; and the heterogeneity by baseline uninsurance—the pattern most directly tied to the coverage mechanism—is nearly exact, with -13.76 (SE 3.58) against BV's -12.40 in high-uninsured counties and -3.49 (SE 4.62) against -3.96 in low-uninsured counties. The pooled estimate with controls, -6.62 (SE 4.14), is smaller in magnitude than BV's -11.36 though the confidence intervals overlap substantially. Two residual differences plausibly account for the gap: our propensity score trims fewer counties than BV's (so our estimand retains more of the high-mortality Southern control counties), and BV's restricted NVSS microdata report small-county death counts exactly where we impute them. We view this reproduction as confirming that the BV design and its key heterogeneity pattern are recoverable from public data, while the pooled point estimate retains some sensitivity to the exact propensity-score implementation. None of this affects the main paper: the comparison in main-paper Table 4 uses BV's published estimate directly, and the weighted-versus-equal SDID comparison is internal to our own panel.

Table 5 closes the loop on sample comparability, comparing population-weighted covariate means in our analysis sample with the values BV publish in their Table 1.

Table 5: Sample comparison with BV2020 Table 1.

	Treated counties		Control counties	
Variable	BV (Treated)	Ours (Treated)	BV (Control)	Ours (Control)
All-cause mortality (per 100,000)	315.17	314.92	359.06	368.41
Unemployment rate	0.09	0.09	0.08	0.08
Poverty rate	0.15	0.15	0.16	0.16
Median income ($10k)	5.95	5.91	5.33	5.09
Uninsured rate	0.19	0.18	0.24	0.24
Obama 2008 vote share[a]	0.46	0.46	0.38	0.38
Obama 2012 vote share[a]	0.43	0.43	0.35	0.35
Democratic governor 2010[a]	0.76	0.76	0.49	0.37

Notes:
Means are weighted by county population aged 20–64 (BV use NVSS microdata; we use the public CDC WONDER panel with suppression-imputed small cells). Mortality rate is the all-cause rate among adults aged 20–64 averaged over 2009–2013.

[a] Unweighted county means. BV's caption describes Table 1 as population-weighted, but their published political-variable values are reproducible only as unweighted county means; we report ours unweighted to match.

F Alternative Approaches to Weighted Estimation

The weighted SDID estimator developed in the main paper is one of three natural strategies for incorporating treated-unit weights. We compare the alternatives here and explain why we adopt the modified-collapsed-form approach.

F.1 Solution 1: Weighted-Average Treated Unit

Collapse all treated units into a single weighted-average row $\bar{Y}_t^w = \sum_j \tilde{\omega}_j Y_{N_0+j,t}$ before running standard SDID with one treated unit. This is simple to implement but has three drawbacks: (i) it loses unit-level heterogeneity that can be diagnostic for understanding the result (cf. Figure 2 of the main paper); (ii) it distorts the cross-sectional dimension, since the algorithm now sees $N_0 + 1$ units rather than $N_0 + N_1$; (iii) it makes bootstrap inference problematic because the single "super-unit" cannot be resampled. Solution 1 is appropriate only when the researcher genuinely has no interest in heterogeneity across treated units.

F.2 Solution 2: Unit-by-Unit SDID

Run a separate SDID for each treated unit j (with that unit as the sole treated unit and all N_0 controls), then compute $\hat{\tau}^w = \sum_j \tilde{\omega}_j \hat{\tau}_j$. This preserves unit-level heterogeneity and allows unit-specific inference, but is computationally intensive (N_1 separate optimizations) and each unit-level SDID has only one treated unit, which can reduce the quality of the synthetic control match for units with idiosyncratic pre-treatment trajectories. The estimator is also less efficient than Solution 3 because the control matching is performed separately for each treated unit rather than jointly against the weighted average.

F.3 Solution 3: Modified Collapsed Form (Our Approach)

Modify the SDID algorithm to incorporate $\tilde{\omega}$ directly into the collapsed form of the outcome matrix. This is theoretically grounded—the synthetic control is constructed to match the researcher's target estimand—computationally efficient (a single optimization on the same-dimensional problem as standard SDID), and admits straightforward variance estimation via the adapted bootstrap, jackknife, and placebo methods. The asymptotic results of Section B carry through with the substitution $\frac{1}{N_1}\mathbf{1} \mapsto \tilde{\omega}$ in the collapsed-form target.

Table 7 reports the comparison. Averaged across the 18 configurations, the RMSE of Solution 3 (0.186) is below that of Solution 2 (0.186) and Solution 1 (0.186). Solution 2 pays the efficiency cost of matching each treated unit separately against the control pool, and the gap widens with the number of treated units. Solution 1 is closest to Solution 3 under homogeneous effects ($\gamma = 0$, mean RMSE 0.186 vs. 0.186), where collapsing treated units discards no estimand-relevant heterogeneity, but it cannot recover unit-level diagnostics and its single-treated-row structure leaves no basis for unit-level resampling inference.

Table 6: Comparison of approaches to incorporating treated-unit weights in SDID.

	Advantages	Disadvantages
Sol. 1: Weighted-average unit	Simple; single SDID run; minimal code change	Loses heterogeneity; collapses to $N_1 = 1$; bootstrap undefined
Sol. 2: Unit-by-unit SDID	Preserves heterogeneity; unit-specific effects available	N_1 optimizations; each uses a single treated unit; less efficient
Sol. 3: Modified collapsed form (ours)	Single coherent optimization; correct estimand; standard inference; asymptotic theory inherits	Requires modified code; intermediate complexity

Table 7: Monte Carlo RMSE of Solutions 1–3 for the weighted ATT.

N_1	γ	HHI	Sol. 1: collapsed unit	Sol. 2: unit-by-unit	Sol. 3: modified form
5	0.0	0.300	0.220	0.219	0.219
5	0.0	0.600	0.305	0.304	0.303
5	0.5	0.300	0.220	0.219	0.219
5	0.5	0.600	0.305	0.304	0.303
5	1.0	0.300	0.220	0.219	0.219
5	1.0	0.600	0.305	0.304	0.303
10	0.0	0.150	0.153	0.153	0.153
10	0.0	0.300	0.208	0.207	0.208
10	0.5	0.150	0.153	0.153	0.153
10	0.5	0.300	0.208	0.207	0.208
10	1.0	0.150	0.153	0.153	0.153
10	1.0	0.300	0.208	0.207	0.208
20	0.0	0.075	0.100	0.100	0.100
20	0.0	0.150	0.131	0.132	0.132
20	0.5	0.075	0.100	0.100	0.100
20	0.5	0.150	0.131	0.132	0.132
20	1.0	0.075	0.100	0.100	0.100
20	1.0	0.150	0.131	0.132	0.132

Notes:
RMSE against the population-weighted target ATT $\tau^{(\tilde{\omega})}$ under the data-generating process of main-paper Section 4 (200 simulations per configuration). Solution 1 collapses the treated units to a single weighted-average row before running standard SDID; Solution 2 averages N_1 single-treated-unit SDID estimates with weights $\tilde{\omega}$; Solution 3 is the modified-collapsed-form estimator adopted in the paper.

G Extensions

This appendix discusses extensions of the weighted SDID framework to settings beyond the simultaneous-adoption single-cohort case considered in the main paper.

G.1 Staggered Treatment Adoption

The main paper assumes simultaneous adoption: all treated units begin treatment at T_0+1. In many applications—including the ACA, where additional states expanded Medicaid after 2014—adoption is staggered. A natural extension applies the weighted SDID estimator within each adoption cohort, then aggregates across cohorts. Let G_g denote the set of units adopting in cohort g at time $T_g + 1$, with cohort weights π_g and within-cohort treated-unit weights $\tilde{\omega}_g \in \Delta^{|G_g|}$. The cohort-aggregated weighted estimand is

$$\tau^{(\pi,\tilde{\omega})} = \sum_g \pi_g \sum_{j \in G_g} \tilde{\omega}_{g,j} \bar{\tau}_{g,j},$$

with the obvious estimator constructed by stacking cohort-specific weighted SDID estimates.

This extension connects to the broader literature on aggregation in staggered DID designs. Callaway and Sant'Anna (2021) and Sun and Abraham (2021) propose group-time-specific ATTs that are then aggregated with researcher-chosen weights; de Chaisemartin and D'Haultfoeuille (2020) emphasize that conventional TWFE can produce negative weights on some group-time cells. Weighted SDID can be viewed as providing analogous flexibility for the synthetic-control component: the researcher chooses how to weight units within each cohort, and the synthetic control is constructed to match that weighted target. A formal treatment of consistency and asymptotic normality under staggered adoption with cohort-level and within-cohort weighting is left to future work.

G.2 Time-Varying Period Weights

The standard SDID estimator averages post-treatment periods uniformly via the $\frac{1}{T_1}\mathbf{1}_{T_1}$ vector in the bilinear form. This can be generalized to researcher-specified period weights $\tilde{\pi} \in \Delta^{T_1}$, useful when:

- short-run vs. long-run effects differ and the researcher wishes to focus on a specific horizon;
- exposure varies across post-treatment periods (e.g., partial implementation in the first year);
- discounting is appropriate for welfare calculations.

Period weights enter the collapsed form through the last column of $\tilde{Y}^w$ and also affect time-weight estimation, since $\hat{\lambda}^w$ is chosen to match pre-treatment periods to the $\tilde{\pi}$-weighted post-treatment average. The asymptotic results of Section B extend naturally provided $\tilde{\pi}$ also satisfies a no-dominant-period analog of W2 — a condition that requires $T_1 \to \infty$ and is therefore of limited practical relevance in short-post-period applications like ours ($T_1 = 4$); in practice, bootstrap inference remains the tool of choice.

G.3 Clustered Standard Errors

In many applications of SDID—including our ACA application—the unit of observation differs from the level of treatment assignment. Counties are the panel units, but Medicaid expansion is a state-level policy. Errors are likely correlated within states, violating the implicit unit-level independence assumed by the bootstrap, jackknife, and placebo variance estimators of Arkhangelsky et al. (2021). Section B (Proposition 3) establishes consistency of the cluster bootstrap under the no-dominant-cluster condition (CB1). The implementation, following Clarke et al. (2023), proceeds as follows:

1. Draw clusters with replacement from the set of all clusters;
2. Include all units within each drawn cluster;
3. Subset the fixed control weights $\hat{\omega}^w$ for the control units appearing in the draw, and renormalize;
4. Subset and renormalize the treated-unit weights $\tilde{\omega}$ for the treated units appearing;
5. Re-estimate $\hat{\tau}^w$ on the resampled data using the subsetted weights.

The cluster jackknife proceeds analogously, leaving out one cluster at a time. The variance formula uses K (the number of clusters) rather than N:

$$\widehat{V}_w^{\mathrm{cjk}} = \frac{K-1}{K} \sum_{k=1}^{K} \left(\hat{\tau}^{w,(-k)} - \bar{\tau}^{w,(\cdot)} \right)^2 .$$

The cluster bootstrap and jackknife are accessed via the software API as `vcov(estimate, method = "bootstrap", cluster = state_fips)` or by passing `cluster` directly to `synthdid_estimate_weighted()`, which stores it for automatic use by `vcov()`; passing `cluster = NULL` to `vcov()` overrides a stored cluster and forces unit-level resampling. The event-study decomposition follows the same convention: `synthdid_event_study()` defaults to the estimate's stored cluster, so its bootstrap confidence bands are cluster-robust whenever the estimate carries one (as in main-paper Figure 1 and Figure E.1, which use the state-clustered bootstrap). In the main paper's ACA application, every cell of Table 2 reports both the county-level bootstrap and the state-clustered bootstrap standard error, with inference based on the latter since treatment is assigned at the state level; the county-level bootstrap likely understates true uncertainty (see the "When the Asymptotics Fail" discussion in Section B).

References

Arkhangelsky, D., Athey, S., Hirshberg, D.A., Imbens, G.W., Wager, S., 2021. Synthetic difference in differences. American Economic Review 111, 4088–4118.

Belloni, A., Chernozhukov, V., Hansen, C., 2014. Inference on treatment effects after selection among high-dimensional controls. Review of Economic Studies 81, 608–650.

Borgschulte, M., Vogler, J., 2020. Did the ACA Medicaid expansion save lives? Journal of Health Economics 72, 102333.

Callaway, B., Sant'Anna, P.H., 2021. Difference-in-differences with multiple time periods. Journal of Econometrics 225, 200–230.

Cameron, A.C., Gelbach, J.B., Miller, D.L., 2008. Bootstrap-based improvements for inference with clustered errors. Review of Economics and Statistics 90, 414–427.

Cattaneo, M.D., Feng, Y., Titiunik, R., 2021. Prediction intervals for synthetic control methods. Journal of the American Statistical Association 116, 1865–1880.

Clarke, D., Pailanir, D., Athey, S., Imbens, G., 2023. Synthetic difference-in-differences estimation. IZA Discussion Paper No. 15907 .

de Chaisemartin, C., D'Haultfoeuille, X., 2020. Two-way fixed effects estimators with heterogeneous treatment effects. American Economic Review 110, 2964–2996.

Politis, D.N., Romano, J.P., Wolf, M., 1999. Subsampling. Springer.

Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. Journal of Econometrics 225, 175–199.